

Bonsai 8

User manual - 2 October 2026 - 0.4.0 candidate

Two songs. Four stems each. One physical player.

Website: [Bonsai 8 / Virtual SP-1](#)

Private development repository: [AWD-tech/bonsai-8](#)

1. Start here

This manual describes **Bonsai 8**, the new name for the custom SP-1 Dual Deck firmware. Its controls differ from the original SP-1 firmware and the upstream Tape Looper. Physical controls below are checked against the current source; new site library, flashing, effects and standalone recording features are identified as candidate work.

Source candidate: Bonsai 8 0.4.0. Last confirmed installed firmware: SP-1 Dual Deck 0.3.5. The installed diagnostic build identifies itself as `sp1-dual-deck-0.3.5`. The new candidate has not been confirmed flashed or hardware-verified. No player was connected for the 2 October checks, and hardware testing is deferred at the user's request. Renaming the repository or opening the site does not update your player.

The existing four-song library was verified after an earlier application update. USB audio and the website's changing waveform were established on the test Mac. Later tests gave smooth playback with four audible stems, but full eight-stem playback and simultaneous USB monitoring still produced underruns. Further mixer optimisation is in the candidate; host results are not a hardware guarantee. See `HARDWARE_TEST.md` for the exact measured conditions.

The firmware loads two songs into **Deck A** and **Deck B**. Each song has four stems. Both decks can play together, but the physical faders and track buttons control only the selected deck. Tapping FUNCTION switches which deck you control; the other keeps its current mix and transport state.

In this library, the stem order is **1 Vocals, 2 Drums, 3 Bass, 4 Other**. Other contains the remaining instruments. A newly recorded A+B mix is different: the complete stereo mix occupies stem 1 only, not four newly separated stems. The device does not recognize instruments by sound: stem order comes from the order used when uploading.

For ordinary playback:

1. Hold FUNCTION for about 1.5 seconds to power on.
2. Deck A starts selected, with the first occupied song slot. Deck B loads the second occupied slot. Both start paused.
3. Keep the master level low initially. Move Deck A's faders through their stored levels to pick them up, then set the mix.
4. Tap and release PLAY to start the selected song. Tap again to pause.
5. Tap and release FUNCTION to switch to the other deck.

At a fresh start, Deck A stem levels are 100%, Deck B stem levels are 0%, both speeds are 1x, and master volume is 25%. Fader pickup prevents an immediate jump when the physical fader and stored level differ.

Installed 0.3.5 has no SFX mode. The Bonsai 8 candidate adds per-stem filter, echo and reverb, plus standalone recording. Section 6 explains their candidate controls and limits.

2. Song 2 vocals over song 1 instruments

This walkthrough starts from a fresh power-on with songs in slots 1 and 2. If you are already mixing, choose those songs on the appropriate decks first and check their existing mutes and levels.

Set up Deck A: song 1 instruments

1. Check that Deck A is selected: the first of the four small side status LEDs indicates A. PLAY and FUNCTION are separate side buttons; the four track buttons belong to the four stems.
2. If necessary, hold FUNCTION and tap the rocker in the next/previous direction to select song 1. Release both. Changing songs stops and cues the selected deck.
3. Tap Track 1 once to mute song 1's vocals. From a fresh start it is unmuted, so one tap turns it off. If it is already muted, leave it muted.
4. For faders 2, 3 and 4, move each fully to the maximum end first to pick up its initial 100% level. Then lower them to the desired instrumental balance. A muted vocal fader can remain anywhere.
5. Tap PLAY. Song 1's drums, bass and other instruments should now play.

Set up Deck B: song 2 vocals

6. Tap FUNCTION briefly to select Deck B. The fourth small status LED indicates B. Deck A continues playing.
7. Select song 2 with FUNCTION plus a short rocker tap if necessary. Deck B starts with song 2 after a fresh boot when slots 1 and 2 are occupied.
8. Move all four faders fully to the minimum end. This picks up Deck B's initial zero levels. Keep faders 2, 3 and 4 at zero. Raise fader 1 for the vocals, and make sure Track 1 is unmuted. You can also mute Tracks 2-4 to prevent accidentally raising their instruments later.
9. Tap PLAY at the musical point where you want the vocals to enter. You now hear **A: song 1 instruments + B: song 2 vocals**. Adjust Deck B's first fader to balance the vocal.

Change the vocal pitch over the other song's drums: keep B selected and move the rocker without FUNCTION. Only B changes speed/pitch; A stays unchanged. This changes all of B's stems together, so keep its non-vocal stems muted or at zero. Pitch and speed move together: preserving vocal timing while changing pitch is not supported.

To rebalance the instruments, tap FUNCTION back to A, pick up its saved fader levels, and adjust them. To stop only the extra vocal, select B and tap PLAY, mute its Track 1, or lower its first fader after pickup.

To start both from the beginning: select A and hold PLAY for at least 0.45 seconds to cue it; select B and do the same. Then hold FUNCTION and tap PLAY once, releasing both. This starts both decks together. It does **not** align their beats or compensate for different intros.

3. Complete physical control reference

Short taps act on release. Hold PLAY or a track button for less than 0.45 seconds when you intend a tap. Press FUNCTION first when using a FUNCTION combination, and perform the second action promptly: holding FUNCTION alone for 1.5 seconds powers off.

Control	What it does
Hold FUNCTION about 1.5 seconds	Power on or power off.
Tap FUNCTION	Switch the selected deck between A and B.
Move a fader	Change that stem's level on the selected deck, after pickup.
Tap Track 1, 2, 3 or 4	Mute/unmute that stem on the selected deck.
Hold one track button at least 0.45 seconds	Temporarily solo that stem on the selected deck. Release to restore the previous mute pattern.
Tap PLAY	Play/pause the selected deck.
Hold PLAY at least 0.45 seconds	Cue the selected song to its beginning and leave it paused.
FUNCTION + tap PLAY	Start both decks if either is paused; pause both if both are playing.
Rocker, faster/forward direction	Increase selected-deck speed by approximately 1% per step.
Rocker, slower/back direction	Decrease selected-deck speed by approximately 1% per step.
FUNCTION + rocker forward	Load the next occupied song on the selected deck, paused at its beginning.
FUNCTION + rocker back	Load the previous occupied song on the selected deck, paused at its beginning.
Volume + / -	Raise/lower the master output for both decks together.
FUNCTION + both volume buttons, held 1 second	Candidate: start/stop standalone recording in the first empty slot. Release both volume buttons before repeating.
Hold FUNCTION + one track button	Candidate: edit that stem's effect with its fader and rocker; release keeps the setting.
Hold Track 1 + Track 4 for 3 seconds	Enter the firmware bootloader, including from charging standby. Playback stops.

Holding a volume button or the rocker without FUNCTION repeats its adjustment after about half a second. FUNCTION plus the rocker changes songs; tap to move one slot at a time. The candidate also repeats browsing while the rocker stays held. Installed 0.3.5 requires separate presses. When FUNCTION plus a track button is held for effect editing, rocker presses choose the effect instead of changing songs.

Solo affects only the selected deck. It does not silence the other deck and does not raise a fader that is at zero. Multiple track-button combinations do not provide additional documented performance functions; Track 1 + Track 4 is reserved for recovery.

Cue and song changes keep the deck's existing speed, gains, mutes and candidate per-stem effect settings. They do not reset its mix. A normal power cycle resets session settings to the startup defaults.

4. Fader pickup, lights and song selection

Browse any song while the other deck plays

Deck A and B are independent selections. They are not limited to adjacent songs: the first and second occupied slots are only the startup defaults.

1. Leave A playing and tap FUNCTION to select B.
2. Hold FUNCTION, then briefly tap the rocker in either direction to choose the next or previous occupied song.
3. Release the rocker before each further tap for deliberate one-song selection. Installed 0.3.5 requires those separate taps; the candidate additionally repeats browsing while held.
4. Repeat to browse all occupied slots; empty slots are skipped and selection wraps around. Release FUNCTION.
5. The newly chosen B song is paused at its beginning. Keep its saved levels/mutes in mind, then tap PLAY to try it over A.

Changing B does not replace A. To browse A while B plays, reverse the deck roles. Both decks may also load the same slot. The candidate site's library selector is intended to select a slot directly for either deck; installed 0.3.5 requires the physical combination above. Loading another song is a storage operation, and uninterrupted transitions under full eight-stem load still need hardware verification.

Fader pickup

Each deck remembers its own four levels during the session. When you switch decks, the fader must reach or cross that deck's saved level before it changes the sound. This prevents the second deck from jumping to the first deck's physical fader positions.

A blinking unmuted track LED indicates pickup is waiting. Move the fader slowly through the saved level until it takes control. If you do not know that level, sweep to one end and then the other; once pickup occurs, further movement changes the audio, so do this with the master low if needed. A muted or missing stem's LED is off even when pickup is waiting.

Four small side status LEDs

Number these four lamps 1 through 4 in order along the side, excluding the separate small dark opening.

Normal display	Meaning
Status LED 1 on	Deck A is selected for the faders/buttons.
Status LED 2 on	Deck A's transport is set to playing.
Status LED 3 on	Deck B's transport is set to playing.
Status LED 4 on	Deck B is selected for the faders/buttons.

Both middle lamps on means both decks are set to playing. A playing lamp alone does not prove audible output: that deck can have zero levels or muted stems.

While FUNCTION is held, these lamps show the selected song slot instead. A steady lamp gives its position within a group of four; a blinking lamp gives the group. Slots 1-4 use group lamp 1, 5-8 lamp 2, 9-12 lamp 3, and 13-16 lamp 4. If the position lamp and group lamp coincide, the steady light masks the blink. For an exact slot number, use the connected website's deck readout.

Four track LEDs

An illuminated track LED means that stem exists and is unmuted on the selected deck. It is **not** an audio-level meter and can stay lit with a zero fader or paused song. Pickup makes an otherwise lit lamp blink. During file transfer all four track lamps blink. Candidate effect editing uses side lamps 1, 2 or 3 to show filter, echo or reverb. Candidate recording overrides the side lamps with a slow blink; draining/pending save uses a faster blink. After saving, normal deck indicators return. Flashing outer side lamps indicate a recording failure; do not assume the take was saved. Alternating side-lamp pairs indicate storage is not available in a recognized format.

Song selection skips empty slots and wraps at the end of the library. Each deck can select any occupied slot, including a song also selected on the other deck. Changing one deck's song leaves the other deck's transport command unchanged; uninterrupted transition performance still needs sustained hardware testing.

5. Timing, loops and getting a good mix

The four stems within a song share a playhead. They stay together when you pause, cue or change that deck's speed. The two decks have independent playheads and speeds. Songs repeat from their beginning when they reach their uploaded length; the uploader pads shorter stems with silence to that song's longest stem.

The primary and secondary songs do not automatically sync to each other's beat or key. There is no master/follower sync setting, beat grid, automatic BPM/key detection, sync button, quantized launch, loop-length control or arbitrary on-device seek. The four stems belonging to one song stay aligned with each other because they share its source timeline; that is different from aligning two different songs. Start the second song by ear. For a restart, hold PLAY to cue that deck and then tap PLAY at the desired moment. This restarts the whole song, not a selected section.

You can adjust volume and pitch while mixing. The four faders adjust the four stem volumes of the selected deck, while the other deck keeps its settings. The physical volume buttons adjust the master for both decks together; there is no separate deck-wide volume knob.

The rocker changes the selected deck's tape-style speed between **0.5x and 1.25x**. Pitch changes with speed. It does not preserve pitch or musical key. If you know both BPMs, the approximate speed ratio for matching B to A is A's BPM divided by B's BPM; use the rocker to approach that value. Matching BPM does not align the first beat, fix drifting performances or match musical keys. Pitch changes apply to every audible stem on that deck together. There is no independent per-stem pitch control, semitone/key selector or pitch-preserving time stretch. For vocals alone, isolate the vocal on B before changing B's speed/pitch; the vocal's timing will change too.

There is no single crossfader. Fade by adjusting the stems on each deck and switching decks as needed. Start with modest levels, then increase the physical master. If the combined sound distorts, lower stem levels and master. The firmware has fixed mixing headroom, but eight loud stems can still clip.

Inserting headphones is intended to mute the built-in speaker automatically. Headphones receive the same combined stereo output; there is no separate headphone cue bus for privately previewing Deck B.

6. Candidate effects and standalone recording

These features are implemented in the **Bonsai 8 0.4.0 candidate**, with software tests. They have not been flashed or checked on the physical test player. Installed 0.3.5 does not have these gestures. Full eight-stem playback already needs further hardware testing; effects and simultaneous recording add more work and are not a verified cure for choppiness.

One effect per stem

1. Select Deck A or B with a **FUNCTION** tap.
2. Hold **FUNCTION** and the desired track button together. Keep them held while editing.
3. Tap the rocker to cycle between **filter**, **echo** and **reverb**. Side status LEDs 1, 2 and 3 identify those choices.
4. Move that stem's fader through the saved effect amount to pick it up, then adjust it. On a new effect, move to minimum first because its amount starts at zero. In this gesture the fader adjusts the effect, not stem volume.
5. Release the buttons. The effect stays active at that amount. The fader returns to stem-volume control with pickup, preventing its new physical position from jumping the saved volume.

Amount zero bypasses the effect. To remove an effect, use the same gesture and bring its amount to zero. Only one effect is selected for each of the eight deck/stem positions. These settings stay after button release and song changes during the session; they are not saved across power cycles. The gesture does not mute or solo the stem on release.

The **filter** is a stereo low-pass filter: increasing the amount removes more high frequencies. **Echo** adds a fixed 250 ms repeat. **Reverb** adds a compact, dark tail. Echo and reverb retain stereo dry sound but use a mono wet path sampled at 4 kHz, so their processed sound has much less high-frequency detail than the original. They are intentionally limited effects, not full-band studio processors.

Echo timing is **not beat-synced**. There is no tempo tap, BPM-linked delay, key matching or automatic sync hidden in this layer. Distortion, stutter/slicer and reverse remain absent. The older upstream Tape Looper guide does not describe this control scheme.

Record the mix without a computer

The candidate records the actual combined, post-master stereo output of A and B, including audible effects and live level/pitch changes. It saves that mix as one stereo P14S stem in the **first empty song slot**. It does not overwrite an occupied song or split the finished mix back into its original eight stems. A compatible library and an empty slot are required.

1. Prepare and start your mix. No browser connection is required.
2. Press FUNCTION and both volume buttons together for about **one second**, then release. The side lamps blink slowly while recording.
3. Perform the mix. The usual faders, mutes, deck switch, speed and effects remain available. Song changes and library transfers are blocked while recording or waiting to save.
4. Use FUNCTION plus both volume buttons for one second again, then release to stop recording. The remaining audio drains to storage. The take then waits to be saved while either deck is still playing; faster side-lamp blinking marks this state.
5. **Pause both decks.** If both are playing, FUNCTION + PLAY pauses them together; otherwise select and pause the remaining playing deck. Both must be paused, not merely muted.
6. Leave the player powered while it flushes the storage cache and reads back the whole take before publishing it. Final saving can take time; transport changes are blocked during this stage. Normal deck lamps return after success. Do not disconnect power or enter another workflow before save completion.
7. Choose the new song slot to play the recorded mix. Its complete sound is on **Track 1**. Raise and unmute that stem as needed. Connect later and use the site's **On your player > Export** action to download it.

The length limit is approximately eight minutes at the stored 24 kHz rate. At the limit, recording stops and follows the same drain/save process. Buffer overflow or storage/verification failure marks the take failed and prevents normal publication; a failed take is not a successful recording. Other song entries are preserved. The inherited storage format has no journal, so power loss during metadata writes is still a risk.

This is a new candidate storage workflow. On-device start/stop gestures, eight-stem-plus-effects write throughput, long takes, power-off recovery and export must still be tested on the hardware. Host tests check encoding, ring ownership, length bounds and metadata preservation, not audible quality or flash durability.

7. What the website is

Open <https://placid-shrine-2h5k.here.now/>.

The **Bonsai 8 companion site / Virtual SP-1** is a companion browser instrument with a rotatable 3D model, music separation, a four-stem browser mixer, stem export and a USB monitor for this firmware. here.now hosts the static site. It is not the firmware running inside the physical player.

Use it without hardware

Choose **Drop a song here** to select audio, or try the studio demo. The site accepts supported WAV, MP3, M4A and FLAC files up to 200 MiB and 10 minutes. Actual decoding depends on the browser. Separation produces Vocals, Drums, Bass and Other locally using a downloaded model. The first separation downloads roughly 158 MiB and can take several minutes. Music is processed on your computer; the site does not send it to a separation server.

Use **Have stems?** to import prepared stems. The browser mixer supports faders, mute/solo, master, speed/pitch and seeking. Drag the 3D player to rotate it or use Camera view to see its sides and bottom. Its geometry uses the approved proportions; Model tools have been removed.

Export stems .zip exports the separated/imported stem audio as WAVs. It is not a recording of your live performance or a two-song mixdown. The existing verified upload workflow is to unzip the WAVs and use the desktop uploader. Direct site-to-device upload is candidate work and must be identified by the connected firmware's supported capabilities.

The website's local instrument and physical firmware have different controls and limits. A gesture on the unconnected model is not an authoritative guide to a physical firmware button combination.

8. Connect, mirror and listen through the site

Use a desktop browser with Web Serial support, such as Chrome or Edge. The player must be powered on in its normal application, **not bootloader mode**. Close any uploader, flashing utility or other serial monitor first; only one connection can own the serial port.

1. Connect a USB data cable and power on the SP-1.
2. Open the website and choose **Connect SP-1**.
3. Select the **Bonsai 8 / Dual Deck** firmware mode offered by the site, click Connect, and choose the matching serial device in the browser's chooser. Installed older firmware may still use the SP-1 Dual Deck name.
4. Play a song and move physical controls. The site shows the selected deck, song number, mixer settings and both deck time/progress readouts.
5. With firmware 0.3, the mirror also receives physical fader positions, button holds/releases and the actual duty values of all eight LEDs.
6. If the operating system recognizes the player's named USB audio input (**SP-1 Dual Deck** on the installed diagnostic build; **Bonsai 8** on a matching renamed build), choose **Listen here** and allow audio-input access. Stop listening to stop computer monitoring; this does not pause the physical player.

On installed 0.3.5, the mirror is read-only: the model's local faders do not set physical gains or pitch. Physical controls drive the physical mix. The candidate library commands are separate actions for choosing a song, uploading or deleting it; the candidate updater is separate again. Do not assume a library selector makes every 3D control a remote hardware control. Connecting does not start an unrelated song previously loaded into the browser.

USB audio is the player's combined post-master stereo mix, not eight separate USB stem channels. Physical master changes therefore affect the captured level. The website's ordinary Master slider belongs to its local browser mixer; it is not a separate USB-monitor volume control. Use the computer output volume to adjust listening level locally.

Current verification limits

Earlier firmware negotiated mirroring and advancing song timers. A complete side-by-side comparison of every physical button/light is still pending. The mirror deliberately buffers about 60 ms; USB and screen refresh add more delay. Screen brightness approximates physical LED brightness. Bootloader and disconnected/off states cannot provide a live mirror.

The test Mac has recognized a stereo 48 kHz **SP_1 Dual Deck** input since 0.3.1, and the site has captured non-silent device audio. The installed diagnostic build is 0.3.5. Capture relays the actual combined device output, so physical playback dropouts can also be heard through the site. Sustained full-eight-stem playback plus capture has not yet passed. Do not interpret a moving timer as proof of clean audio. No computer microphone is connected to playback as a substitute for the player input.

The time readouts show each song's source position and reset at its loop boundary. Changing speed changes how quickly those positions advance. They are not beat markers. The live waveform, when capture works, describes the combined audio. Audio and visuals have different transport delays; sample-accurate alignment is not claimed.

9. Prepare and upload songs

The physical player plays prepared stems; it does not separate a mixed song on-device. You can use the website to make stems, or export them from an audio editor. Use four files with the same starting point, in this order: **vocals, drums, bass, other**. Keep leading silence needed for alignment.

The library holds up to 16 song slots, with four stems per song. The current uploader limits songs to approximately eight minutes, even though the website accepts up to ten minutes. It converts uploads to 24 kHz stereo P14S; the device outputs a 48 kHz stereo mix. The player also recognizes compatible P16M mono files, but not the old raw-PCM or ADPCM library used by earlier firmware.

Candidate site library controls

The Bonsai 8 candidate work adds a connected-device library view with occupied slots, selection of any loaded song for A or B, deletion of one chosen song, and upload of prepared stems. The **On your player** panel provides **Load A**, **Load B**, **Export** and **Delete** actions. These controls require compatible firmware; they are not verified capabilities of installed 0.3.5. The original four songs do not need re-uploading for the rename.

Inspect the slot and song before deleting or replacing it. Deletion removes its library entry and makes the slot reusable; it is not secure erasure of the underlying audio. Confirm the chosen slot before a delete or replacement. These actions are designed to keep other songs intact and refresh the library afterward. Uploads/transfers pause playback. Keep source stems on your computer, and wait for verified transfer completion before unplugging. End-to-end device checks are still required before these flows can be called verified.

Desktop uploader

Run these commands from the firmware repository. The examples use the project's Python environment; uploading also needs ffmpeg. Disconnect the website serial connection first.

```
.build-env/python/bin/python tools/sp1.py ports
.build-env/python/bin/python tools/sp1.py status
.build-env/python/bin/python tools/sp1.py list
```

Upload to an empty slot, substituting your four actual file paths:

```
.build-env/python/bin/python tools/sp1.py upload --slot 5 \
  vocals.wav drums.wav bass.wav other.wav
```

Quote file paths containing spaces. To target a specific device when discovery is ambiguous, put `--port /dev/cu.usbmodemXXXX` before the command name. Use the actual port reported by `ports`.

An occupied slot is refused by default. Adding `--replace` explicitly replaces that slot; interruption can leave that song empty or incomplete. Uploading pauses both decks. The uploader verifies written sectors and flushes storage before completion. Wait for success before reconnecting the website.

The four songs already uploaded to this player do not need initialization or re-uploading for a firmware update. The destructive `init --erase-all-songs` command replaces the library index. Do not use it to solve a playback, connection or audio-input problem. Unknown storage is preserved and write-locked until an explicit initialization is authorized.

Keep the original source stems on your computer. Song audio persists across power cycles; deck selection, levels, mutes, speed and play positions do not.

10. Troubleshooting

Symptom	What to check
Song 2 appears to play but is silent	Deck B starts at zero. Select B, move its vocal fader to minimum for pickup, raise it, unmute that stem and check master.
A fader seems unresponsive	It is probably waiting for pickup. Cross the selected deck's saved level. Switching decks re-arms pickup.
Vocals from both songs are audible	Select A and mute its Track 1. Select B and confirm only its vocal is audible.
Solo does not isolate one song	Solo only affects the selected deck. Pause/mute the other deck separately if you want complete isolation.
Both songs are playing but drift apart or clash in key	There is no automatic beat/key sync. Match speed and launch timing manually; speed also changes pitch. Musical key is not detected or corrected.
Only the next song seems available	Select the intended deck, then use separate FUNCTION + rocker taps. Release the rocker between taps, and use either direction. All occupied slots are searchable.
A song starts at the beginning after PLAY	PLAY was held long enough to cue. Use a short tap for pause/resume.
FUNCTION turns the unit off	It was held alone for about 1.5 seconds. Tap briefly to switch decks; press the other control promptly for a combination.
Both middle status lamps light but there is silence	Check mutes, fader pickup, master, loaded stems and headphones. A playing flag is not an audio meter.
Site cannot connect	Power on normally, use a data cable and a Web Serial browser, and close other serial owners. Bootloader mode is for flashing.
Site connects but Listen here fails	Check whether the OS lists the player's named audio input, then check browser permission and selected input. Older installed builds identify as SP-1 Dual Deck; renamed candidate builds identify as Bonsai 8.
Site audio echoes against the player	The two output paths have different delay. Monitor through one listening path or reduce one path's listening volume.
Status LEDs alternate in pairs	Storage is unavailable or its format is unsupported. Inspect diagnostics; do not initialize an existing library blindly.
Playback glitches	Hardware testing has recorded playback-buffer underruns. Save a status log with both deck settings; two-deck performance remains under investigation.
No SFX controls can be found	Installed 0.3.5 has none. The candidate uses FUNCTION + a track button held together; that stem's fader sets amount and rocker chooses the effect.
Recording stopped but no new song appears	Pause both decks and wait for draining, cache flush and complete readback. Pending or failed takes are not published as ready songs.
A recorded mix is silent except on Track 1	The mix is saved as one stereo stem on Track 1; the other three stems are intentionally empty.

To collect a three-minute diagnostic log without an active website serial connection:

```
.build-env/python/bin/python tools/sp1.py status --seconds 180
```

11. Updates, development and credits

Development lives at [AWD-tech/bonsai-8](https://github.com/AWD-tech/bonsai-8), a private repository for source changes, issues and pull requests. See README.md for build instructions and HARDWARE_TEST.md for measured results. Use a feature branch and a pull request for future changes; keep uploaded audio, build environments and account credentials out of commits.

Candidate browser updater

The candidate site's **Install Bonsai 8** action opens its application-only flashing workflow. It must validate the matching firmware image/manifest before erasing application pages, and it must not initialize or upload the music library. A displayed progress bar or a completed download is not proof that firmware was installed. Bootloader acknowledgement and a subsequent runtime version check are separate verification steps. This browser flow still needs its own hardware test; installed firmware remains 0.3.5 until an update and runtime check establish otherwise.

Firmware updates and library uploads are different operations. A firmware update writes the application; it does not intentionally reformat the song library. For an update, disconnect the site's serial connection, hold Track 1 + Track 4 for three seconds with USB connected, and use the release's matching updater and binary. Afterward, hold FUNCTION about 1.5 seconds if the application needs powering on. Reconnect normally and verify the reported version, controls and stored songs.

The root `sp1_looper.bin` and the older Tape Looper documents are upstream references, not the custom Bonsai 8 release. Do not select a binary merely because its filename contains SP-1. Keep the application binary, its checksum/manifest and the updater version together. Candidate `bonsai-8-0.4.0` has not been established as flashed by this manual.

This firmware is based on `chattock/sp1-tape-looper` at commit

`44ba1ecbec6c844dba7f47eacee94c53af8ab10d`. It retains the hardware initialization and recovery foundations while replacing the active audio/control path with a two-deck mixer. Original MIT license and attribution are retained. It is a custom community build, not an official SP-1 firmware release.

Control instructions were checked against `firmware/src/dual_firmware.inc` and `firmware/src/dual_engine.c`. Website behavior was checked against the companion app and its hardware/audio modules. Implemented behavior, software tests and confirmed hardware results are distinguished throughout this manual.